Pay Attention to Network: Reliability-Aware Spatial-Temporal-Frequential Scheduling for TSN-WiFi Networks

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Abstract

Time-Sensitive Networking (TSN) is a pivotal technology in providing deterministic and reliable communications in the wired domain. Complementarily, Wi-Fi Multi-link Operation (MLO) extends the TSN capabilities over the wireless domain to further increase the throughput and reliability for wired-wireless-integrated scenarios such as smart factories in industrial automation. To incorporate TSN and Wi-Fi MLO seamlessly, cross-domain flow scheduling is of vital importance in reliable control message transmission. Existing approaches leverage Deep Reinforcement Learning (DRL) to adapt to the dynamic wireless channel quality. However, the traditional multilayer perception (MLP)-based DRL model fails to integrate the input features of all the flows, resulting in severe underfitting issues and scheduling ineffectiveness. In this paper, we observe that the mutual dependencies among flows over the network resources are analogical to the contextual dependencies of tokens over the lexical sequence. Leveraging this insight, we exploit the self-attention mechanism in Natural Language Processing (NLP) to learn the network-wise dependencies among different flows. Theoretically, to provide an analytical guide to TSN-WiFi flow scheduling, we present a global scheduling abstraction to describe the key resource constraints of the cross-domain flow scheduling problem. Following that, we propose the load-aware TSN scheduling algorithm and the attention-based DRL scheduling algorithm to allocate spatial-temporal-frequencial resources for flows adhering to the above constraints. Experiments in various settings validate the effectiveness of the proposed method, which enhances the reliability by 12.09% compared with the state-of-the-art method.

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CCS Concepts

• **Networks** → **Link-layer protocols**; **Packet scheduling**.

Keywords

Time-Sensitive Networking, Wi-Fi, Flow scheduling, Deep Reinforcement Learning

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1 Introduction

Time-Sensitive Networking (TSN) being developed under the IEEE 802.1 Task Group is a pivotal technology to provide deterministic transmission, bounded low latency, and high reliability as a foundation for time-critical applications [30]. TSN has been successful in wired networks but lacks the flexibility and scalability required by emerging applications that extend the requirement for time-critical communication and high reliability from the wired domain to more distributed, flexible, and mobile environments [24]. For example, smart factories, Autonomous Guided Vehicles (AGV), and remote offices are applications that would greatly benefit from easy reconfigurability and smooth user experience through untethered connections [16].

Hopefully, the advances in the latest wireless technologies IEEE 802.11be Extremely High Throughput (EHT) [14] enable the extension of TSN capabilities over wireless seeking to further increase the throughput, reduce the end-to-end latency, and increase the reliability of communications. To achieve this, Multi-link Operation (MLO) is considered a disruptive feature update as it promotes the use of multiple wireless interfaces to allow concurrent data transmissions in Access Points (APs) and Stations (STAs) with tri-band capabilities (i.e., 2.4 GHz, 5 GHz, and 6 GHz). Consequently, incorporating TSN with Wi-Fi MLO would be a significant solution for seamless operation across wired and wireless domains [3].

To accelerate the TSN-WiFi integration, IEEE 802.1 [13] makes standardization efforts by proposing potential scheduling collaboration between 802.1Qbv [1] in TSN and 802.11be in Wi-Fi MLO

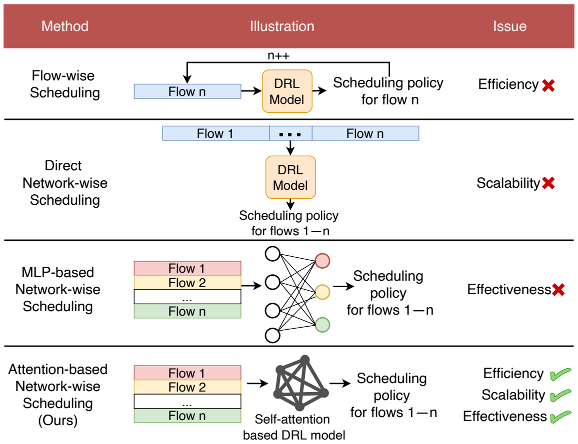


Figure 1: The comparison of different DRL algorithms.

to pre-allocate space, time, and frequency resources for critical traffic. Despite being insightful, it fails to provide a solution to the cross-domain scheduling problem. Many researchers investigate the potential solutions for scheduling management architecture design [22] and scheduling function validation for single devices [7]. However, existing studies are limited to either high-level modular design or impractical assumptions of static channel conditions, inapplicable to realistic large-scale networks.

Recently, Deep Reinforcement Learning (DRL) has emerged as a promising approach for flow scheduling due to its self-exploiting capability, which can well adapt to dynamic wireless channel information and adjust its scheduling policy accordingly. Meanwhile, cross-domain scheduling is a multi-dimensional function that takes the network resource information and the channel state information as input and outputs the spatial, temporal, and frequential resources allocated to each flow. Such a problem can be formulated as an optimal control problem of a Markov decision process (MDP), which can be solved by DRL [26].

Previous DRL algorithms suffer from different issues as Fig. 1 shows. Flow-wise scheduling [12] schedules each flow sequentially, causing a long execution time because the number of times for inference is equal to the number of flows. Direct network-wise scheduling [11] solves the above problem by taking the direct concatenation of all the flow information as input and generating the scheduling result for all the flows at once. However, when the flow number changes, the dimension of the DRL model will be incompatible to the input feature, which causes the scalability issue. MLP-based network-wise scheduling [36] tackles the scalability issue by input reshaping with the information of each flow being a vector of the whole batch to keep the input dimension constant. Although the MLP-based model could generate the scheduling result for each flow, the feed-forward process for each flow vector is independent without integrating the information from other flows. This causes underfitting without fully exploiting the input information and will thus lead to the ineffectiveness of the scheduling result generation.

Diving deep into the scheduling problem, we observe that there are mutual dependencies among flows. When a flow is scheduled, it will occupy certain resources which will potentially influence the schedulability of other flows due to the resource limitation. Thus, it is of vital importance for the DRL model to learn the dependencies among different flows over the resource availability and further estimate which flows to schedule to optimize the problem. In Natural Language Processing (NLP), self-attention is utilized to learn the contextual relationships by enabling each token in an input sequence to "attend" to every other token in the same sequence [29]. Interestingly, each flow information in the whole network is analogical to each token information in the whole context, which gives us a hint that we can exploit the self-attention mechanism in NLP to learn the network-wise dependencies among different flows. As Fig. 1 shows, we take self-attention as the building block of the DRL model to learn the flow dependencies and generate the scheduling results for all the flows. Moreover, we design an actor-critic Deep Deterministic Policy Gradient (DDPG) algorithm to guide the learning process as it can achieve great time efficiency with its policy and long-term reward both being approximated by two Neural Networks (NNs). Thus we leverage the attention-based DDPG to conduct the cross-domain scheduling problem satisfying efficiency, scalability, and effectiveness simultaneously.

To realize that, we further consider the following challenges. **(1) Resource dimension distinction between TSN and Wi-Fi domains.** TSN scheduling primarily relies on space-time resources to determine the routing and timing of flow transmission. On the other hand, Wi-Fi scheduling manages flows by assigning the specific frequency band and airtime for them. This resource dimension mismatch poses difficulties in cross-domain coordination. To address it, mathematically, we set up cross-domain scheduling constraints to regulate the resource selection in different dimensions. Methodologically, we design TSN load-aware scheduling algorithm to determine the route and time schedule in TSN domain to leave more free space for Wi-Fi domain. **(2) Continuous and large decision space in Wi-Fi domain scheduling.** Wi-Fi scheduling tasks include frequency band selection and airtime search that request multi-step decisions. Meanwhile, airtime is a continuous value, causing an infinite space for decision-making. This results in search inefficiency and convergence difficulty. To tackle this issue, we convert the problem into a ternary search problem for each flow to determine the selected band, which discretizes the decision variable and reduces the search space. Following that, we develop the lower bound for the airtime to regulate the time allocation. **(3) Ineffective raw feature values for DRL model training.** The raw values observed directly from the environment are noninformative and in high variability, which causes model instability and degraded scheduling performance. To cope with that, we utilize the prior knowledge of wireless channel transmission to get the implicit features of bands and incorporate them into the input features of the DRL model to accelerate its learning process.

In this paper, we present the cross-domain spatial-temporal-frequential scheduling scheme for TSN-WiFi networks including architecture design, system model formulation, and scheduling algorithm design. We compare it with other state-of-the-art methods. The results from experiments in different settings show that we acquire a 12.09%/20.49% reliability improvement over SLCI (Single Link Less Congested Interface)/Random methods. The contributions are as follows:

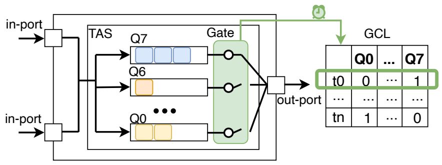


Figure 2: TSN TAS switch.

• We introduce the first cross-domain TSN-WiFi scheduling scheme comprehensively solving the multi-dimensional spatial-temporal-frequential resource allocation for flows in TSN-WiFi hybrid networks. It fills the research gap of scheduling in large-scale TSN over Wi-Fi scenarios.
• We provide a global scheduling abstraction to describe the key resource constraints in both TSN and Wi-Fi domains for flow scheduling. Based on it, we propose a load-aware scheduling algorithm and an attention-based DDPG scheduling algorithm for TSN and Wi-Fi domains, respectively.
• We conduct experiments on realistic topologies and different flow volumes to validate the effectiveness and adaptiveness of the proposed method. The high reliability it achieves demonstrates its robustness.

2 Related Work

Existing TSN-WiFi scheduling can be categorized into three types based on which domain the scheduling policy applies to, TSN scheduling, Wi-Fi MLO scheduling, and TSN-WiFi integrated scheduling.

2.1 TSN Scheduling

Conventionally, the scheduling algorithms model the TSN scheduling problem as a standard Satisfiability Modulo Theories (SMT) [4, 23] or Integer Linear Programming (ILP) problem [6, 31], which are then solved using a solver. Solver-based methods generate the optimal schedule by exploring the whole search space, yielding exponential time complexity. Heuristic algorithms save execution time by exploiting heuristics to search in a limited space. For example, Tabu heuristics is used in No-wait Packet Scheduling Problem (NW-PSP) [5] and Injection Time Planning (ITP) [32] to calculate TSN schedules. Moreover, Greedy Randomized Adaptive Search Procedure (GRASP) [9] is also used in TSN scheduling algorithm. With the advancement of DRL, numerous methods for addressing scheduling problems have emerged [12, 34].

2.2 WiFi MLO Scheduling

Bellalta *et al.* [19–21] propose three approaches for flow allocation to multiple interfaces. One is the Single Link Less Congested Interface (SLCI) allocating a new incoming flow to the least congested interface. The second is Multi-Link Same Load to All Interfaces (MLSA) allocating an equal fraction of the incoming flow on all the interfaces. The last is the Multi-Link Congestion-Aware Load Balancing at Flow Arrivals (MCAA) approach, which distributes the incoming flow among different enabled interfaces in proportion to the amount of free channel airtime on each interface. Since the segmentation of a flow will cause extra dividing and verification effort for both APs and STAs, we do not consider these segmentation-based methods (MLSA and MCAA) in our work. LFTA [17] considers the co-existence with non-MLO devices by allocating flows till only a certain limit so as to spare channel capacity exclusively for single-link non-MLO devices.

2.3 TSN-WiFi Scheduling

Gilson *et al.* [22] present a modular controller architecture to provide control over TSN and Wi-Fi domains. However, it only identifies the required building blocks to support cross-domain scheduling but not the system model and scheduling method. Scholars from Intel [7, 27] investigate the interoperability between devices in TSN domain and Wi-Fi domain by realizing time synchronization, TSN flow scheduling, and data path redundancy in both domains. However, these efforts are limited to proof-of-concept validation and prototype implementation without theoretical analysis and scalability guidance for practical wired/wireless scenarios. IEEE 802.11be standard [3] also proposes collaboration of TSN and Wi-Fi, and specifies deterministic transmission in Wi-Fi. Whereas, the collaboration method and architecture still remain unsolved. In this work, we try to tackle cross-domain TSN-WiFi scheduling issue from the aspects of architecture, system model, and algorithm design.

3 Preliminary

3.1 TSN and Wi-Fi MLO Model

3.1.1 TSN.

As Fig. 2 shows, all the TSN switches and host devices are synchronized with a global clock time at the nanosecond level specified in IEEE 802.1AS [2]. With the presence of an accurate global time, TSN switches can allocate transmission time for flows within the system using a Gate Control List (GCL), which can accurately manage the timing of flow transmission for each queue. The operating logic of GCL follows the traffic shaping mechanism Time-Aware Shaper (TAS) in IEEE 802.1Qbv [1]. In TAS, GCL operates at the granularity of per packet. Implementing the TAS mechanism necessitates meticulous consideration of the sequence and timing of each data packet's entry and exit from the queue. As shown in Fig. 2, the time slot t_0 in the TSN switch is specifically reserved for the flow in Queue 7.

3.1.2 Wi-Fi MLO.

To increase the Wi-Fi throughput while reducing the end-to-end delay and improving the reliability of communications for time-sensitive applications, MLO is considered a main candidate feature in IEEE 802.11be EHT [14]. It promotes the use of multiple wireless interfaces to allow concurrent data transmissions in APs and STAs with tri-band capabilities, i.e., 2.4 GHz, 5 GHz, and 6 GHz bands. To achieve that, the standard proposes Multi-Link Devices (MLD) that refer to AP or STA devices with multiple wireless interfaces (links). As Fig. 3 shows, within each MLD, the MAC layer is divided into two sub-layers, Upper MAC (U-MAC) and Lower MAC (L-MAC) which conduct different sub-layer functionalities.

U-MAC: Common management functions for all links such as setup, association, and traffic management take place in this layer. To handle incoming flows, the traffic manager operates at the U-MAC layer to distribute new incoming flows among different interfaces so that they can be transmitted over multiple links.

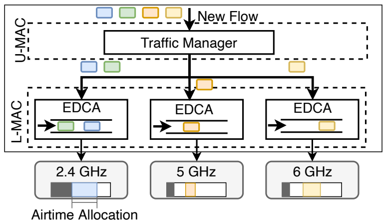


Figure 3: Wi-Fi MLD.

L-MAC: The L-MAC is independent for each interface and is in charge of link-specific functionalities like channel access. It considers the channel load information and calculates the airtime that can be allocated to each flow. Each interface keeps track of its own Enhanced Distributed Channel Access (EDCA) queue to hold the traffic until the corresponding transmission time.

3.2 Scheduling Problem and Formulation

Here we consider a flow transmission scheme as Fig. 4 shows. We only consider the control flows from end stations in the TSN domain to end stations in the Wi-Fi domain. In each cycle time T_{cycle}, there are n flows that need to be transmitted, which form a set F. To reflect the dynamic feature caused by customized orders or emergency commands in the industrial field, the departure times of these flows O^{start} from their source devices vary in each cycle. The flow transmitted cross-domain can be defined as follows.

$$\begin{aligned} &\forall f_a \in F,\ a \in [0,\ n-1],\ a \in Z, \\ &f_a = \{src, dst, size, O^{start}, band, reliability\}, \end{aligned} \tag{1}$$

where src is the source device in TSN domain, dst is the destination device in Wi-Fi domain, $size$ is the frame length of the flow, O^{start} is the start time on src, $band$ is the selected band among multi-link interfaces in Wi-Fi domain, and $reliability$ is the reliability of the flow, which is affected by the channel quality of the selected $band$.

As flows transmit cross domain in every cycle, they shall meet the transmission constraints per cycle in each domain to avoid conflict and packet loss.

3.2.1 *TSN Domain.*

The topology of the TSN network is modeled as a directed graph $G = \{V, E\}$. The set of vertices V is composed of the set of end devices ES and the set of switches SW, denoted as $V = ES \cup SW$. End devices serve as the source and destination of flows while switches only forward flows. The set of edges E represents TSN links. The network supports full-duplex transmission, and thus the TSN links are directed. For example, two vertices $v_m, v_n \in E$ determine two TSN links $[v_m, v_n]$ and $[v_n, v_m]$. For simplicity, the TSN link can also be denoted as TL.

For flow f in the TSN domain, it is transmitted following its path $P_f = [TL_0, TL_1, ..., TL_{m-1}]$, which contains m hops. P_f is an ordered sequence of TSN links and $TL_i (0 \le i \le m-1)$ is the ith TSN link within the path.

Frame Constraint: The offset on each TSN link has to be greater than or equal to the start time from the source and the transmission window has to fit within the cycle.

$$\forall f \in F, \forall TL \in P_f, O_f^{start} \le O_f^{TL} < T_{cycle}, \tag{2}$$

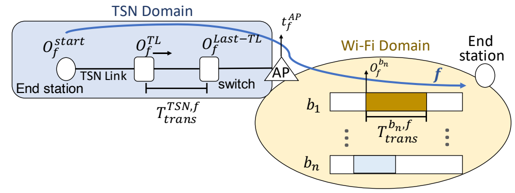


Figure 4: Flow transmission scheme.

where O_f^{TL} is the offset on the TSN link TL of f.

Transmission Constraint: The offsets of a flow must follow the sequential order along the path.

$$\forall f \in F,\ \forall TL_i, TL_{i+1} \in P_f, O_f^{TL_{i+1}} \ge O_f^{TL_i} + T_{trans}^{TSN,f}. \tag{3}$$

Here $T_{trans}^{TSN,f}$ is the transmission time of f on each TSN link calculated with $\frac{f.size}{DR_{TSN}}$, where $f.size$ and DR_{TSN} are flow size of f and TSN data rate, respectively.

Link Constraint: Two flows routed through the same link cannot overlap in the time domain.

$$\begin{aligned} &\forall f_a, f_b \in F, \forall TL \in P_{f_a} \cap P_{f_b}, \\ &(O_{f_a}^{TL} + T_{trans}^{TSN,f_a} \le O_{f_b}^{TL}) \vee (O_{f_b}^{TL} + T_{trans}^{TSN,f_b} \le O_{f_a}^{TL}). \end{aligned} \tag{4}$$

Deterministic Constraint: Link-sharing flows need to be scheduled so their arrival times are far apart to avoid interleaving in the queue.

$$\forall TL_i \in P_{f_a} \cap P_{f_b}, (O_{f_a}^{TL_i} \le O_{f_b}^{TL_{i-1,f_b}}) \vee (O_{f_b}^{TL_i} \le O_{f_a}^{TL_{i-1,f_a}}), \tag{5}$$

where TL_{i-1,f_b} and TL_{i-1,f_a} are the preceding TSN link of TL_i on the path of f_b and f_a respectively.

3.2.2 *Wi-Fi Domain.*

In Wi-Fi MLO, the available band set is $B = \{b_1, b_2, b_3\}$, which stands for 2.4 GHz, 5 GHz, and 6 GHz respectively. Each band $b_n \in B$ has its reliability $b_n.reliability$ calculated as Eq. (6) [35]. For each flow f, it selects a band b_n to transmit in Wi-Fi domain, which is denoted as $b_n(f) = 1$. Then a duration of airtime window is assigned to f, which starts at offset time $O_f^{b_n}$.

$$b_n.reliability = 1 - f_Q\left(\frac{-f.size\ln 2 + \Delta(t)BW_{b_n}\ln[1 + \mathrm{SNR_n(t)}]}{\sqrt{\Delta(t)BW_{b_n}V_n(t)}}\right), \tag{6}$$

where f_Q is Q-function, $f.size$ is the size of f in bits, $\Delta(t)$ is the transmission time of f in the t-th cycle time, BW_{b_n} is the bandwidth of b_n, $SNR_n(t)$ is the Signal-to-Noise Ratio of b_n in the t-th cycle time, and $V_n(t)$ is the channel dispersion defined as $V_n(t) = 1 - 1/[1 + SNR_n(t)]^2$.

Frame Constraint: The transmission time in Wi-Fi domain has to be within the cycle.

$$\forall f \in F, b_n(f) = 1, O_f^{b_n} + T_{trans}^{b_n,f} \le T_{cycle}, \tag{7}$$

where $T_{trans}^{b_n,f}$ is the transmission time of f on band b_n calculated with $\frac{f.size}{DR_{b_n}}$. DR_{b_n} can be calculated with the Shannon Formula.

$$DR_{b_n}(t) = BW_{b_n}\log_2(1 + SNR_n(t)) \tag{8}$$

Band Constraint: Two flows transmitted on the same band cannot overlap in the time domain.

$$\forall b_n \in B, \forall f_a, f_b, b_n(f_a) = b_n(f_b) = 1, \\ (O_{f_a}^{b_n} + T_{trans}^{b_n,f_a} \le O_{f_b}^{b_n}) \vee (O_{f_b}^{b_n} + T_{trans}^{b_n,f_b} \le O_{f_a}^{b_n}). \tag{9}$$

Occupancy Constraint: Each flow can at most choose one band to transmit.

$$\forall f \in F, \sum_{n=1}^{3} b_n(f) \le 1 \tag{10}$$

3.2.3 Cross Domain.

Cross-domain Constraint: The flow can only start transmitting in Wi-Fi domain after it accomplishes its transmission in TSN domain.

$$\forall f \in F, b_n(f) = 1, O_f^{b_n} \ge O_f^{Last-TL} + T_{trans}^{TSN,f}, \tag{11}$$

where $O_f^{Last-TL}$ is the offset time on the last TSN link of f.

FIFO Constraint: If flows are assigned the same band, their transmission should follow the sequence of arrival at AP.

$$\forall b_n \in B, \forall f_a, f_b, b_n(f_a) = b_n(f_b) = 1, \\ (t_{f_a}^{AP} > t_{f_b}^{AP} \wedge O_{f_a}^{b_n} > O_{f_b}^{b_n}) \vee (t_{f_a}^{AP} < t_{f_b}^{AP} \wedge O_{f_a}^{b_n} < O_{f_b}^{b_n}), \tag{12}$$

where t_f^{AP} is the arrival time of f at AP computed with $t_f^{AP} = O_f^{Last-TL} + T_{trans}^{TSN,f}$.

If a flow meets the above constraints in a cycle time, it is successfully scheduled and denoted as $S(f) = 1$. Since the indeterminacy of flow transmission is caused mainly by noise interference in the wireless domain, we denote the reliability of flow as the reliability of its selected band in Wi-Fi.

$$\forall f \in F, b_n(f) = 1, f.reliability = b_n.reliability. \tag{13}$$

The optimization objective can be defined as follows.

Optimization Objective: We aim to optimize the overall reliability of the flows in each cycle.

$$\textbf{Max} \sum_{f \in F} f.reliability \times S(f). \tag{14}$$

Decision Variables: The decision variables of the problem are in three categories: 1) **Spatial:** the path of each flow in TSN domain $P_f = [TL_0, ..., TL_{m-1}]$, 2) **Temporal:** the offsets on each hop in TSN domain O_f^{TL} and Wi-Fi band $O_f^{b_n}$, and 3) **Frequential:** the band selection in Wi-Fi domain b_n.

4 Method

The TSN-WiFi cross-domain scheduling problem is NP-hard, meaning there is no polynomial-time algorithm that can generate the optimal solution [18]. To address the above problem both effectively and efficiently, we first propose a scheduling architecture for the TSN-WiFi hybrid network. Following that, we introduce the scheduling method in TSN and Wi-Fi domains respectively.

Algorithm 1: Load-aware TSN flow scheduling

Input: Flow set F, Network N
Output: Path P and Offset O^{TL}
for $f \in F$ **do**
  Get the path set S_p and bottleneck resource of each path based on N
  Select the path P with the maximum bottleneck resource for f
  for $TL \in P$ **do**
    for $TimeSlot \in T_{cycle}$ **do**
      Check the Constraints of Eq. (2), Eq. (3), Eq. (4), and Eq. (5)
      if *Meet Constraints* **then**
        O_f^{TL}=$TimeSlot$, **break**
      end
    end
  end
end

4.1 Overview

As Fig. 5 shows, there are two parts in the whole network: TSN domain and Wi-Fi domain. Flows transmit from TSN domain to wireless hosts in Wi-Fi domain. To achieve reliable communication, flow scheduling strategies are embedded in both domains. For TSN domain, the load-aware TSN scheduling module is deployed in Central Network Configuration (CNC), which is a centralized component in TSN responsible for collecting the flow information, conducting scheduling task, and configuring the switches with GCL. For Wi-Fi domain, the scheduling module is deployed in AP, which is the bridge between TSN and Wi-Fi, and it can collect both the flow information and Wi-Fi wireless channel parameters for scheduling. To adapt to the dynamism of channel quality and flow features, we utilize the "explore and exploit" nature of DRL in the scheduling module and propose an attention-based DDPG method to allocate frequence and time resources to flows in every cycle.

4.2 TSN Load-Aware Scheduling

The network throughput for flows is a key performance indicator in TSN domain. There exists a bucket effect in TSN routing and scheduling that the schedulability of a flow is determined by the minimum available resource along the flow path. Taking Fig. 6 as an example, the number of overall occupied time slots is 4 in both paths. However, in path $[l_1, l_2]$, the minimum number of available time slots is 0 on link l_2, which is the bottleneck of this path and thus results in scheduling failure. In contrast, the minimum number of available time slots is 2 in path $[l_3, l_4]$, meaning the link with the minimum resource is sufficient for the flow and the flow is successfully scheduled in this path.

Taking this insight, we design a load-aware scheduling algorithm to improve the schedulability of flows and the load balance level of the network resources. The pseudocode is presented in Alg. 1, where we refer to the minimum available link resource along the path as the *bottleneck resource*. Then we assign the earliest possible time slot for the flow on each link to leave more time for scheduling in Wi-Fi domain.

4.3 Wi-Fi DDPG Scheduling Design

The DDPG model of Wi-Fi scheduling is shown in Fig. 7, which follows an actor-critic structure. The main part of both actor and critic is the attention-based model at the bottom of Fig. 7. It is

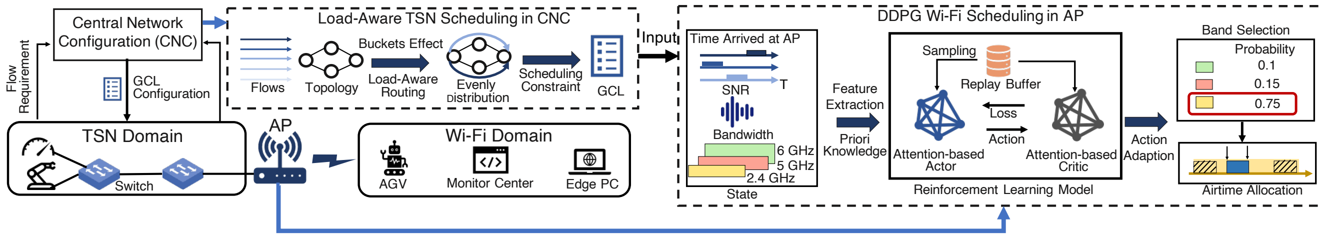


Figure 5: Overview of the scheduling architecture.

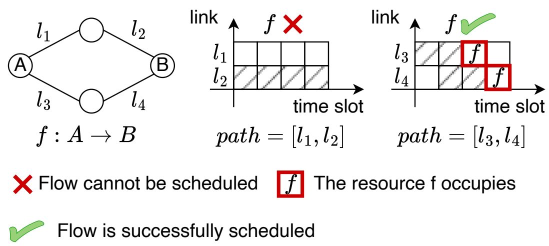


Figure 6: Illustration of bucket effect in TSN scheduling. The dashed boxes are previously occupied time slots.

the combination of two networks: 1) Self-attention network. It captures the relationships among input vectors irrespective of their distance in the sequence [25]. Inspired by this insight, we leverage it to capture the dependencies among flows; 2) Residual network. There is a shortcut path where the input is added to the output of the self-attention network. This helps gradients flow through the network during backpropagation, mitigating the vanishing gradient problem [33].

4.3.1 State Modeling.

The state observed by the DRL model in AP comes from two parts: 1) the scheduling results in TSN parts. This is utilized to check the cross-domain constraint in Eq. (11) and compensate for the large delay in TSN domain by determining an appropriate offset on the available band in Wi-Fi domain; 2) the channel information of three bands in Wi-Fi domain. Methodologically, there are two considerations in the state feature extraction.

Form of Network-Wise State Feature: Different from flow-wise inference, the network-wise state includes the information of all the flows. However, integrating all the flow information and channel parameters into one single vector would cause variable input dimensions inapplicable to the fixed DRL network. To tackle this, we utilize vector v_i with a fixed length to represent the feature of each flow f_i and formalize a state matrix with all the flow vectors $State_A = [v_1^T, v_2^T, ...v_n^T]$, where n is the number of flows.

Content of Knowledge-Assisted State Feature: Flow vector v_i consists of three parts: (1) the arrival time of f_i at AP; (2) the SNR of three Wi-Fi channels; (3) the bandwidth of three Wi-Fi channels, which is denoted as $v_i = [t^i_{ap}, SNR_1, SNR_2, SNR_3, BW_1, BW_2, BW_3]^T$. As these numbers are in different orders of magnitude which may cause an imbalance over different features of the model, we normalize them into the same scale $v_i = [\frac{t^i_{ap}}{T_{cycle}}, \frac{SNR_1}{SNR_{max}}, \frac{SNR_2}{SNR_{max}}, \frac{SNR_3}{SNR_{max}}, \frac{BW_1}{BW_{max}}, \frac{BW_2}{BW_{max}}, \frac{BW_3}{BW_{max}}]^T$. Although SNR and BW are the raw information we can get directly from the Wi-Fi channel, there are

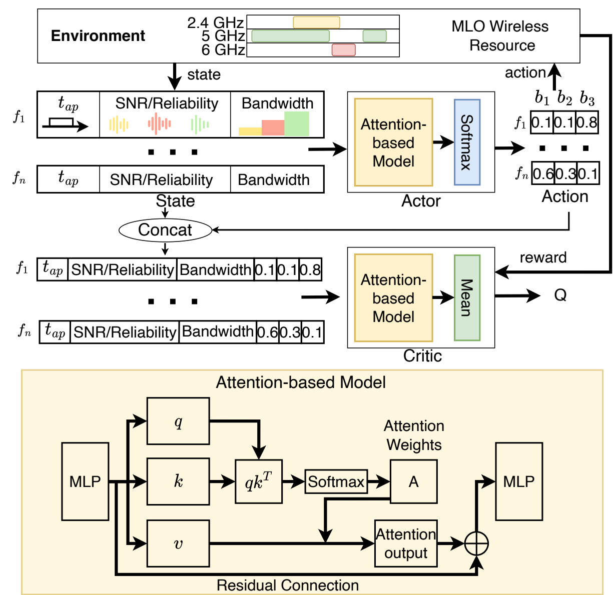


Figure 7: The structure of DDPG model. In order to distinguish the Query in the attention network from the Q value in the RL model, the Query, Key and Value of the attention network are denoted by q, k and v respectively.

some issues with v_i: 1) The variation of $\frac{SNR_k}{SNR_{max}} (k = 1, 2, 3)$ in scale is still very large with the minimum value being 0.0387 and the maximum value being 1. 2) The transformation from raw information SNR and BW to the optimization goal $F.reliability$ is non-linear and complicated for the model to learn. To tackle these issues, we incorporate band reliability into the state feature with the prior knowledge in Eq. (6). The scale of band reliability is [0.42,1], which is compressed and benefits the learning of the model with the implicit band information. As a result, the content of v_i is $[\frac{t^i_{ap}}{T_{cycle}}, \frac{SNR_1}{SNR_{max}}, \frac{SNR_2}{SNR_{max}}, \frac{SNR_3}{SNR_{max}}, b_1.reliability, b_2.reliability, b_3.reliability, \frac{BW_1}{BW_{max}}, \frac{BW_2}{BW_{max}}, \frac{BW_3}{BW_{max}}]^T$.

The input of the actor network is denoted as a matrix $[v_1^T, v_2^T, ..., v_n^T] \in R^{n \times 10}$, where n is the number of flows. The input of the critic network can be denoted as a matrix $[w_1^T, w_2^T, ...w_n^T]$, where $w_i = \text{concat}(v_i || a_i)$. a_i is the i-th action vector generated by the actor network elaborated in Sec. 4.3.2.

4.3.2 Action Modeling.

The original scheduling tasks in Wi-Fi include band selection and offset time search, which request multi-step action. Meanwhile, the offset time is a continuous value, causing an infinite action space. A reinforcement learning algorithm converges as the number of visits to each state-action pair approaches infinity [28]. When the action is multi-step and its search space is large, finding the optimal scheduling plan is nearly impossible in practice. To overcome this, we simplify the action step and discretize it in the following two aspects.

Action Generation: If a flow f selects a band b_n, its offset time is lower bounded by the following inequation constrained by Eq. (11) and Eq. (12).

$$O_f^{b_n} = \max\{t_f^{AP}, O_{\overline{f}}^{b_n} + T_{trans}^{b_n,\overline{f}}\}, \overline{f} \in F, t_{\overline{f}}^{AP} < t_f^{AP} \tag{15}$$

As the transmission in Wi-Fi is constrained by T_{cycle} in Eq. (7), we can take the lower bound as the offset of f to guarantee its earliest transmission. Thus, the actor only needs to determine which band to select for each flow. We define the eventual action of the actor in cycle t as $\hat{a}(t) = [b_{f_1}(t), ..., b_{f_n}(t)]$, where $b_{f_i}(t) \in \{0, 1, 2\} (i = 1, ..., n)$ representing the band selection from $\{b_1, b_2, b_3\}$ of f_i. Thus the number of possible actions is 3^n, which is much smaller than that of the original infinite action space.

As the last layer of the actor network is softmax layer, the direct output of the action is the probability of each band selection for each flow $a(t) = [S_1^T, ...S_n^T] \in R^{n\times 3}$. The probability of band selection for f_i is $a_i(t) = S_i^T = [p_{i,b_1}, p_{i,b_2}, p_{i,b_3}]$. Following that, we can sample the band for each flow according to this distribution and generate the eventual action $\hat{a}(t)$.

Action Adaptation: As we need both band selection and offset time on that for the scheduling plan in Wi-Fi, we shall conduct action adaptation from $\hat{a}(t)$ according to Eq. (15). Thus, the offset time of f on the selected band b_n can be determined as follows:

$$O_f^{b_n} = \begin{cases} O_{\overline{f}}^{b_n} + T_{trans}^{b_n,\overline{f}}, \text{ if } \exists \overline{f}, b_n(\overline{f}) = 1, O_{\overline{f}}^{b_n} + T_{trans}^{b_n,\overline{f}} > t_f^{AP}, \\ t_f^{AP}, \text{ otherwise.} \end{cases} \tag{16}$$

4.3.3 Model Updating.

Updating Algorithm: As shown in Fig. 5, DDPG is an actor-critic DRL algorithm, where the actor and the critic are two NNs that determine the action in a given state and evaluate the long-term reward of the state-action pair. Given a system state, the action to be executed is obtained from the actor,

$$a(t) = \mu(s(t)|\theta^{\mu}), \tag{17}$$

where $s(t)$ and a(t) are the state and action in the t-th cycle, and $\mu(\cdot|\theta^{\mu})$ is the actor and θ^{μ} is the parameters of the actor.

Given $s(t)$ and $a(t)$, the long-term reward is estimated by a state-action value function by the critic $Q(s(t), a(t)|\theta^{Q})$, where θ^{Q} is the parameters of the critic.

In each cycle t, the model observes the current state $s(t)$ and generates an action according to Eq. (17). After taking a certain action, the model observes the instantaneous reward $r(t)$ and the state in the (t+1)-th cycle. The transition $< s(t), a(t), r(t), s(t{+}1) >$ is stored in a replay buffer BF.

To update the model, a batch of transitions are selected from BF and used as the training samples to optimize the parameters of the NNs. To optimize the parameters of the critic, we use Bellman equation [28],

$$Q(s(t_i), a(t_i)) = E[r(t_i) + \gamma Q(s(t_i+1), \mu(s(t_i+1)|\theta^{\mu})|s(t_i), a(t_i)]. \tag{18}$$

Algorithm 2: Wi-Fi DDPG scheduling

Initialize the parameters of the NNs, θ^{Q}, θ^{μ}
Initialize the temporal copies of the parameters, $\hat{\theta}^{Q} \leftarrow \theta^{Q}$, $\hat{\theta}^{\mu} \leftarrow \theta^{\mu}$
for episode $m = 1, ...M$ **do**
  Set the system to an initial state
  for cycle $t = (m-1)T + 1, ..., mT$ **do**
    Set O_f^{start} for flows and SNR for bands
    Observe state $s(t)$
    Generate action from $a(t) = \mu(s(t)|\theta^{\mu})$ and execute the action
    Evaluate the reward $r(t)$ from Eq. (21) and observe the next state $s(t+1)$
    Save transition $< s(t), a(t), r(t), s(t+1) >$ in BF
    Select N transitions as a batch of training samples
    Exploit Eq. (19) and Eq. (20) to update $\hat{\theta}^{Q}$ and $\hat{\theta}^{\mu}$
    Update θ^{Q} and θ^{μ}:
      $\theta^{Q} \leftarrow (1-\tau)\theta^{Q} + \tau\hat{\theta}^{Q}$, $\theta^{\mu} \leftarrow (1-\tau)\theta^{\mu} + \tau\hat{\theta}^{\mu}$
  end
end

The realization of the right-hand side of Eq. (18) in the t_i-th cycle is defined as $y(t_i) = r(t_i) + \gamma Q(s(t_i{+}1), \mu(s(t_i{+}1)|\theta^{\mu})|\theta^{Q})$. To obtain an accurate approximation of the long-term reward, the DDPG algorithm minimizes the difference between $y(t_i)$ and $Q(s(t_i), a(t_i)|\theta^{Q})$. Thus the parameters of the critic are optimized by minimizing the following loss function,

$$L(\theta^{Q}) = \frac{1}{N}\sum_{i=1}^{N}[y(t_i) - Q(s(t_i), a(t_i)|\theta^{Q})]^2, \tag{19}$$

where N is the batch size. Since the optimal policy maximizes the state-action value function, the loss function of the actor is

$$L(\theta^{\mu}) = -\frac{1}{N}\sum_{i=1}^{N} Q(s(t_i), \mu(s(t_i)|\theta^{\mu})|\theta^{Q}). \tag{20}$$

The pseudo-code of DDPG is presented in Alg. 2.

Reward Design: According to the optimization goal in Eq. (14), the total instantaneous reward in the t-th cycle is defined as the total reliability of all the flows in this cycle,

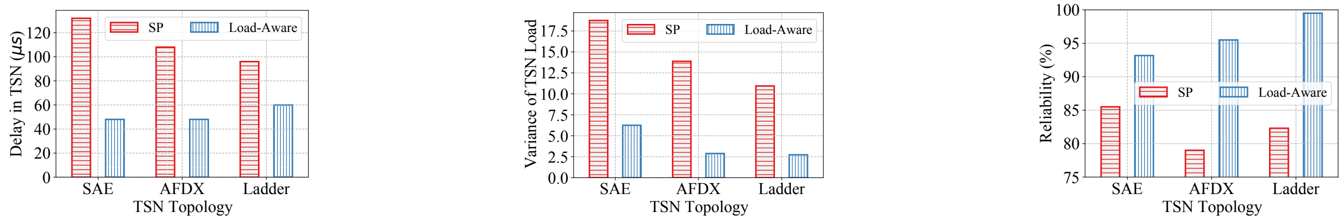
$$r(t) = \sum_{i=1}^{n} f_i.reliability(t). \tag{21}$$

5 Evaluation

5.1 Setup

5.1.1 Network Setting.

To test the scalability and robustness of the proposed algorithm, we conduct experiments on three realistic topologies: SAE (Society of Automotive Engineers) [8], AFDX (Avionics Full-Duplex Switched Ethernet) [36], and Ladder [36]. We set the TSN link data rate as 1 Gbps. The cycle time T_{cycle} is chosen from {5 ms, 10 ms}. The channel bandwidths of the three bands in Wi-Fi {2.4 GHz, 5 GHz, 6 GHz} are set as 20, 80, and 160 MHz, respectively [17]. The SNR of these channels is randomly chose from [-6.18, 6.97]. The



(a) Maximum delay of flows in TSN domain.

(b) Variance of resource distribution in TSN network.

(c) Reliability of flows with different TSN scheduling algorithms.

Figure 8: Performance of TSN scheduling algorithms in different topologies.

length of the time slot in TSN is the transmission time of an MTU (Maximum Transmission Unit)-sized frame (1500 bytes), i.e., 12 μs.

5.1.2 Flow Feature.

To cover different flow volumes, we generate two flow sets {SetA, SetB} with 50, and 100 flows, respectively. The size of each flow is set as one MTU, i.e., 1500 bytes[8]. For each flow, the destination and source are selected randomly from the end stations. The start offset of each flow is randomly selected from 0∼4 time slots.

5.1.3 Hyper-Parameters in DRL.

We try different values and choose the best ones in this section. The learning rates of both actor and critic, and the soft-updating rate are 0.0004. The capacity of the replay buffer is 8000. The batch size of DDPG model updating is 50. The discount factor γ is 0.9. For both actor and critic, the MLP layers before q, k, and v include one input layer and two hidden layers with a dimension of 64. The activation function is `ReLU`. The output layer behind the attention output has a dimension of 3, which is the action size.

5.1.4 Baseline.

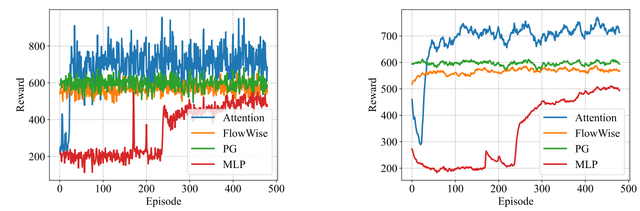
To test the performance of the load-aware scheduling algorithm in TSN, we compare it with the Shortest Path (SP) algorithm [10]. To test the performance of the attention-based DDPG algorithm (Attention) in Wi-Fi, we compare it with the flow-wise DRL algorithm (FlowWise) [12], Policy Gradient algorithm (PG) [36], and MLP-based DRL algorithm (MLP) [15]. To compare our algorithm with non-DRL schedulers, we evaluate the Round-Robin (RR), Earlist-Deadline-First (EDF) [11], SLCI [21], and Random schedulers.

5.2 Results

5.2.1 Performance of TSN Load-Aware Scheduling Algorithm.

Fig. 8a shows the maximum delay of flows in three topologies. The average maximum delay of SP algorithm in three topologies is 112 μs, which is 2.15× that of the load-aware scheduling algorithm. The reason is that the SP algorithm always selects the shortest path for flows without considering the available resources and conflicting conditions in the flow path, causing severe congestion and accumulated delays when flows share the same links in the network. In contrast, the load-aware algorithm takes the bucket effect of scheduling into consideration and especially pays attention to the bottleneck resource on the most resource-constrained link to avoid congestion and large delays.

Fig. 8b shows the variance of resources on the links after scheduling in TSN domain. The average variance in three topologies of the load-aware algorithm is 3.95, reducing the variance of SP algorithm by 3.68×. The reason is that the load-aware algorithm always selects the less-loaded path from the path sets and avoids adding additional flows to the most resource-constrained links. Thus the links are more evenly occupied by flows resulting in a balanced resource distribution.



(a) Training reward.

(b) Reward moving average.

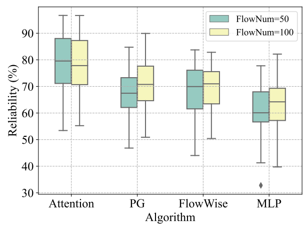
Figure 9: The reward in the training process.

Fig. 8c demonstrates the ultimate reliability of flows after scheduling in both TSN and Wi-Fi domains. It evaluates the effect of TSN scheduling algorithm on our optimization goal. The average reliability with the load-aware algorithm is 96.06%, exceeding that of SP algorithm by 13.78%. The reason is that the extended delays of flows with SP algorithm causes the late arrival time at AP. Thus the available airtime in Wi-Fi band will be shrunk due to the cross-domain constraint in scheduling, which results in the scheduling failure of flows and thus the decrease in the average reliability.

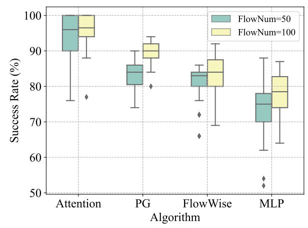
5.2.2 Comparison among Different Wi-Fi Scheduling Algorithms.

The training processes of different scheduling algorithms are shown in Fig. 9. Our attention-based DDPG achieves the highest reward because the attention-based DRL model can learn the dynamic dependencies among flows and adapt to the wireless channel features to figure out the scheduling decisions. The reward of Policy Gradient algorithm oscillates around 600 and fails to get higher because the loss comes from the reward expectation, which tends to be easily affected by the sampled transitions and slow to converge. Flow-wise DRL algorithm fails to learn the relations among different flows and the result is prone to be influenced by the input sequence of flows. MLP-based DRL scheduling gets the least reward because it exploits the information of each flow independently and fails to integrate them, demonstrating the superiority of the attention-based model over the MLP-based model.

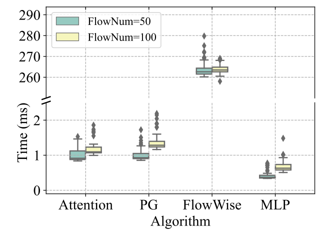
Fig. 10a shows the average reliability of flows scheduled with different algorithms. The average reliability of our proposed algorithms is 78.63% when the flow number is 50, exceeding PG, FlowWise, and MLP by 11.23%, 10.53%, and 18.23%, respectively.



(a) Average reliability of flows.

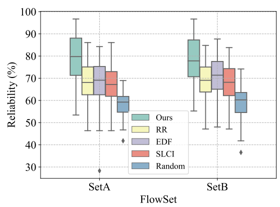


(b) Scheduling success rate.

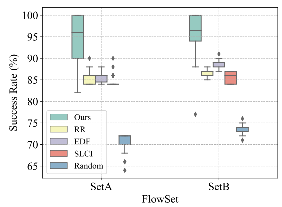


(c) Execution time.

Figure 10: Performance of Wi-Fi scheduling algorithms in different flow volumes.

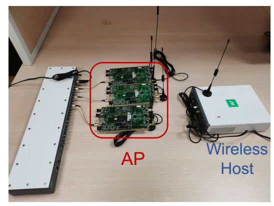


(a) Average reliability of flows with different schedulers.

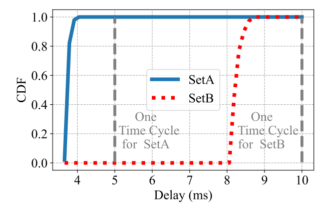


(b) Scheduling success rate with different schedulers.

Figure 11: Performance of Wi-Fi schedulers in Set A and B.



(a) Testbed setup.



(b) Flow delays in Set A and B.

Figure 12: Testbed deployment setup and result.

This is because the attention-based algorithm could dynamically study the relationships among different flows and also the channel parameters to generate the scheduling policy that could increase the overall reliability of the network. The reliability when the flow number is 100 shows the same trends, demonstrating the robustness of the proposed algorithm in different flow volume scenarios. Note that the reliability of the flows with the proposed algorithm ranges from 50% to 100% because the reliability of Wi-Fi bands is in the range of [43%, 99.99%].

Fig. 10b shows the scheduling success rate of flows scheduled with different algorithms. The average success rate of our proposed algorithms is 96% when the flow number is 100, exceeding PG, FlowWise, and MLP by 6.38%, 13.14%, and 18.46%, respectively. This is because the attention-based algorithm could dynamically study the relationships among different flows and also the channel parameters to generate the scheduling policy that could make the resources evenly distributed, triggering the successful scheduling of more flows.

Fig. 10c shows the execution time of the algorithms. When the flow number is 50, the average execution times are 1.04, 1.19, 263.97, and 0.43 ms for the proposed algorithm, PG, FlowWise, and MLP, respectively. The execution time of the proposed algorithm and PG are slightly the same because they share the same DRL model for inference. The flow-wise algorithm is the most time-consuming because it generates the scheduling policy for each flow in sequence that needs to be conducted for the number of times the same as the flow number. The execution time of MLP is the least because the MLP model is lighter weight than the multi-layer attention model, which saves forward time in inference. The execution time of our proposed algorithm is within one cycle time, which is applicable in our scenario.

5.2.3 *Comparison of Existing Schedulers.*

Fig. 11a shows the average reliability of flows with different schedulers. The average reliability of our proposed algorithms is 78.84% when the flow number is 50, exceeding RR, EDF, SLCI and Random by 10.79%, 11.28%, 12.09%, and 20.49%, respectively. This is because the attention-based algorithm could dynamically study the relationships among different flows and also the channel parameters to generate the scheduling policy that could increase the overall reliability of the network. Other schedulers follow the fixed rules and fail to dynamically adapt to the changing wireless quality.

Fig. 11b shows the scheduling success rate of flows with different schedulers. The average success rate of our proposed algorithms is 96% when the flow number is 100, exceeding RR, EDF, SLCI, and Random by 9.5%, 7.2%, 10.4%, and 22.27%, respectively. The scheduling success rate when the flow number is 50 shows the same trends, demonstrating the robustness of the proposed algorithm in different flow sets. This is because the attention-based algorithm could dynamically learn the relationships among different flows and also the channel parameters to generate the scheduling policy that could make the resources evenly distributed, triggering the successful scheduling of more flows.

5.2.4 *Testbed Deployment.*

To verify the schedule calculated by the proposed method, we deploy the scheduling result into USRP B210 to act as the AP responsible for receiving flows from TSN domain and transmitting flows in Wi-Fi domain to wireless hosts (another USRP N210). We configure the USRP to transmit signals in tri-band, i.e., 2.4 GHz (20 MHz), 5 GHz (40 MHz), and 5 GHz (80 MHz) because of the device limitation beyond 5 GHz. Since our algorithm can adapt to different parameter settings, we apply the above values to the DDPG method and perform real scheduling tests of the flow delays in Set A and Set B. Fig. 12b illustrates the cumulative distribution function of flows in different flow sets. The maximum delays of flows in Set A and Set B are 3.934 and 8.626 ms, which are both within their corresponding cycle times of 5 and 10 ms, respectively. The experiment demonstrates that both sets of flows, when scheduled via the proposed method, can ensure a transmission delay within the maximum allowed cycle time and a 100% scheduling success rate.

6 Conclusion

In this work, we introduce the first cross-domain TSN-WiFi scheduling scheme comprehensively solving the multi-dimensional spatial-temporal-frequential resource allocation problem for flows in TSN-WiFi hybrid networks. Theoretically, we provide a global scheduling abstraction to describe the key resource constraints in both TSN and Wi-Fi domains for flow scheduling. Following that, we propose a load-aware scheduling algorithm for TSN domain and a DDPG scheduling algorithm for Wi-Fi domain exploiting the self-attention mechanism in NLP to learn the mutual dependencies among flows over the resource availability. The experimental evaluation demonstrates the effectiveness of the proposed method.

Acknowledgments

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